

Akash Viswanathan

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SUMMARY

U.S. Citizen | Public Trust Cleared | CS & Economics @ Northeastern University

Full-Stack Engineer with 1+ years of experience in high-performance C++ and Python simulation (AFSIM), and AI-driven automation. Proven track record of cutting deployment times by 85% and building RAG systems for complex technical datasets.

EXPERIENCE

Full Stack Engineer Co-op

Stellar Science

August 2025 - December 2025, Albuquerque, NM

- Designed a retrieval-augmented generation system in Python capable of indexing 100k+ AFSIM functions, giving engineers a natural-language interface to explore simulation logic and reducing onboarding and debugging time.
- Developed surrogate AI/ML models (custom neural nets and random forests, 92%+ accuracy) that replaced costly physics simulations, cutting runtime and enabling faster experimentation and analysis.
- Built high-performance C++ modules in ETEM and AFSIM, increasing simulation throughput and enabling larger, more complex defense scenarios to run within operational time constraints.
- Supported cloud modernization aligned with CMMC standards, improving reliability and reducing risk in secure deployment environments that underpin mission workflows.

DevOps Engineer Co-op

State Street/Charles River Development

January 2024 - August 2024, Burlington, MA

- Automated migration of 1,000+ Azure user accounts with Python and PowerShell, cutting provisioning time by >85% and eliminating setup errors through validation and logging.
- Created lifecycle automation scripts that reduced routine IT tickets by 50%, freeing support teams to focus on higher-impact operational tasks.
- Built and deployed CI/CD pipelines for Azure resources, reducing deployment time by 30% and improving release consistency across teams.
- Integrated Azure Monitor with Grafana dashboards to provide real-time visibility into production, reducing incident response time by 40% and improving reliability.

PROJECT

Real-Time Gesture & Posture Recognition System — Python, C++, OpenCV, MediaPipe, NumPy, ONNX

- Built a hybrid real-time computer vision system for hand gesture classification and full-body pose tracking, integrating custom ML models with low-latency inference.
- Collected and labeled 5,000+ hand-gesture samples using MediaPipe and a custom cv2 interface, enabling supervised learning from 2D keypoints.
- Built a NumPy-based neural network and custom KNN classifier, achieving 90%+ accuracy with softmax confidence filtering and top-k prediction.
- Implemented real-time full-body tracking in C++ using an ONNX-exported YOLOv8 pose model and OpenCV DNN, rendering keypoints at ~30 FPS.
- Designed a modular socket-based architecture for cross-language integration, allowing Python gesture predictions to be rendered live in a C++ overlay.

InfraPulse — Python, AWS, Docker, SQLite, Grafana, Terraform

- Built a cloud-based monitoring system to track uptime and response time for 10+ web services, storing health checks in SQLite and visualizing real-time metrics in Grafana dashboards.
- Wrote Python scripts to perform 60+ health checks per hour, logging response time and status for trend analysis and alerting.
- Configured automated email alerts via AWS SES, enabling issue detection and response within <5 minutes of a failure event.
- Containerized the system using Docker, enabling deployment in under 30 seconds across local and cloud environments.
- Used Terraform to provision AWS infrastructure as code, supporting reproducible environments and future scaling.

CourseMatch — SQL, Docker, Appsmith

- Connected a custom SQL database to Appsmith, enabling users to search, filter, and manage 500+ course and section records based on their role (e.g., student, professor, faculty).
- Boosted data access speeds by 30% through query optimization and real-time filtering using Appsmith's dynamic widgets and SQL binding.
- Cut onboarding time for new contributors by 90% via Docker-based containerization, ensuring seamless local development and reproducible deployment.

EDUCATION

Bachelor of Science in Computer Science and Economics

Northeastern University • Boston, MA • May 2026 • 3.7 GPA

- Activities: Teaching Assistant (Java, Python, Kotlin), Northeastern Electric Racing (Software Team), Code4Community
- Relevant Coursework: Algorithms and Data Structures, Object-Oriented Design, Database Design, Artificial Intelligence, Applied Data Analytics, Economic Theories, Statistical Machine Learning

SKILLS

Languages: Python • Java • TypeScript • SQL • C++ • R

AI/ML: TensorFlow • scikit-learn • Pandas • PyTorch • NumPy • HuggingFace Transformers

Frameworks & Tools: React • Flask • Express.js • AWS (SES, S3, Lambda, EC2) • Azure (AZ-900 and AI-900 Certified) • Jenkins • Terraform • Linux